

Physics

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1 Quantum Gravity

1. Overview, Quantum Gravity, and Quantum Theory of the Gravitational Field
2. Semiclassical Gravity Problems and Approaches to Quantum Gravity
3. Covariant Approaches to Quantum Gravity and the Concept of Gravitons
4. Path-integral quantization
5. Parametrized Systems, Relational Dynamics, Particle Models, and Bosonic Strings
6. Parametrized field theories
7. Hamiltonian formulation of general relativity
8. Quantum geometrodynamics
9. Semiclassical approximation
10. Loop Quantum Gravity
11. Quantization of black holes
12. Quantum Cosmology

2 The Early Universe

1. Overview and the expanding Universe
2. Cosmological Redshift
3. Distance Measures and Temporal Evolution of the Scale Factor
4. Horizons and de Sitter space
5. The hot Universe
6. Leptonic Phase and Nucleosynthesis
7. Recombination, Baryogenesis, and Topological Defects
8. Inflation in the Homogeneous Limit: Flatness and Horizon Problems
9. Horizon Problem and Inflationary Scenarios
10. Inflationary Scenarios and Examples of Inflationary Potentials
11. Gravitational instability in Newtonian theory
12. Gravitational Instability in Newtonian Theory and General Relativity
13. Gravitational Instability in General Relativity and Inflation II

3 Quantum Mechanics

Quantum Mechanics 1

1. [Introduction](#)
2. [U\(1\) Representations](#)
3. [Two-State Systems and SU\(2\)](#)
4. [Linear Algebra](#)
5. [Lie Algebras and Lie Algebra Representations](#)
6. [Rotations and Spin in 3 Dimensions](#)
7. [Rotations and Spin 1/2 in a Magnetic Field](#)
8. [Representations of SU\(2\) and SO\(3\)](#)
9. [More Representations of SU\(2\) and SO\(3\)](#)
10. [Tensor Products](#)
11. [Momentum and the Free Particle](#)
12. [Fourier Analysis and the Free Particle](#)
13. [Position and the Free Particle](#)
14. [Heisenberg Group & Schrodinger Representation](#)
15. [Poisson Brackets and the Symplectic Group](#)
16. [Quadratic Polynomials and the Symplectic Group](#)
17. [Hamiltonian Vector Fields and the Moment Map](#)
18. [Quantization](#)
19. [Semi-Direct Products](#)
20. [Representations of the Euclidean Group and the Free Particle](#)
21. [Representations of Semi-Direct Products](#)
22. [Central Potentials and the Hydrogen Atom](#)

- 23. [The Harmonic Oscillator](#)
- 24. [Coherent States](#)
- 25. [Review](#)

Quantum Mechanics 2

1. [Canonical Quantization, Second Semester Overview](#)
2. [Fermionic Oscillators, Weyl and Clifford Algebras](#)
3. [Clifford algebras and Geometry](#)
4. [Anticommuting Variables and Pseudo-Classical Mechanics](#)
5. [Fermionic Quantization and Spinors](#)
6. [Complex Structures and Supersymmetry](#)
7. [Supersymmetric Quantum Mechanics](#)
8. [The Pauli Equation and the Dirac Operator](#)
9. [Lagrangian Methods and the Path Integral](#)
10. [Multi-Particle Systems: Momentum Space](#)
11. [Multi-Particle Systems and Field Quantization](#)
12. [Review of Symmetries and Quadratic Operators](#)
13. [Symmetries and Non-Relativistic QFT](#)
14. [Minkowski Space and the Lorentz Group](#)
15. [Representations of the Lorentz Group](#)
16. [The Poincare Group and its Representations](#)
17. [The Klein-Gordon Equation and Scalar Quantum Fields](#)
18. [Symmetries and Propagator for Scalar Fields](#)
19. [More Symmetries and Propagators for the Scalar Field](#)
20. [U\(1\) Gauge Symmetry and Electromagnetic Fields](#)
21. [Quantization of the Electromagnetic Field](#)
22. [More on Quantization of the Electromagnetic Field](#)
23. [The Dirac Equation and Spin 1/2 Fields](#)
24. [More on the Dirac Equation and Spin 1/2 Fields](#)
25. [The Standard Model](#)

4 Quantum Field Theory

Quantum Field Theory 1

1. Introductory words; using a model of a crystalline solid to discover QFT
2. Discovery of Fock space
3. Scalar field theory with some extra adjectives
4. Quantization of the radiation field by analogy; Noether's theorem and its converse
5. Casimir effect; fields mediate forces
6. Fields mediate forces; Wick rotation and time-ordering
7. Big picture recap; Feynman diagrams in 0+0-dimensional QFT
8. Lorentz invariance and causality
9. Path integral subtlety; propagators; where is single-particle QM in QFT?
10. The S-matrix and Dyson expansion
11. Wick's theorem; time-ordered Green's functions and diagrams
12. The exponentiation of the disconnected diagrams
13. The LSZ reduction formula
14. Observable physics from the S-matrix
15. Observable physics from the retarded Green's function; Group theory
16. Representations of the Lorentz group on fields
17. Spinor Lagrangians
18. The quanta of a spinor field are fermions
19. Scattering of fermions
20. Vector fields and gauge invariance

Quantum Field Theory 2

1. Goals, parable on violation of scale invariance by QM
2. Beta function; a simple example of perturbative renormalization
3. First steps towards quantum corrections to the Coulomb force law
4. Electron self energy; self-energy in ϕ^4 theory
5. Where is the cutoff dependence?
6. QED vertex correction
7. Soft photons
8. Vacuum polarization
9. Spectral density
10. Optical theorem and cutting rules
11. Hadrons with perturbative QCD; Parable on integrating out heavy modes
12. Anderson-Higgs mechanism
13. Interlude on differential forms
14. Gauge fields as connections; fermion path integrals
15. Lattice gauge theory
16. Gauge fixing
17. The QCD beta function
18. Wilsonian RG, briefly
19. Renormalization of composite operators; Callan-Symanzik equation; Introductory words about Effective Field Theory
20. The color of the sky; loops in effective field theory; the Standard Model as an effective field theory

Quantum Field Theory 3

1. Goals; Anomalies
2. Anomalies; emergence of the Dirac equation in 1+1d
3. More anomalies
4. Effective field theory of gravity and of particle physics
5. Effective field theory of superconductors and superfluids
6. Effective field theory of metal
7. Effective field theory of metal, continued
8. Coherent state path integrals for bosons and fermions
9. Path integrals and sums for spins; geometric quantization
10. Coherent state path integrals for spins
11. Topological terms from integrating out fermions; pions
12. More on pions
13. Nucleons as solitons; Field theory of spin systems
14. Field theory of the Neel phase
15. Large N
16. Transverse-field Ising model
17. Field theory of the transverse-field Ising model
18. Superfluid-insulator transition and T-duality
19. Bosonization
20. Holographic duality

5 Relativity and Cosmology

Relativity and Cosmology 1

1. [Overview](#)
2. [A brief history of spacetime](#)
3. Newtonian gravity [A](#) [B](#)
4. Special relativity
 - (a) [Poincaré transformation](#)
 - (b) [Spacetime diagrams and the relativity of simultaneity](#)
 - (c) [Time dilation and Lorentz contraction](#)
 - (d) [Spacetime and four-vectors](#)
 - (e) [Special relativistic mechanics](#)
5. Gravity as geometry
 - (a) [The twin problem in special relativity and Gravitational redshift](#)
 - (b) [Motion in a gravitational field](#)
 - (c) [Newtonian limit \(heuristic derivation\)](#)
 - (d) [Curved spaces](#)
6. Differential geometry I: Classic formulation
 - (a) [Differentiable manifolds, Vector and tensor fields, and The metric tensor](#)
 - (b) [Tensor densities](#)
 - (c) [Riemann normal coordinates](#)
 - (d) [Tensor analysis](#)
 - (e) [The Riemann curvature tensor](#)
 - (f) [Further discussions of the Riemann tensor and the Weyl tensor](#)
 - (g) [Integration and Invariant tensor fields and Lie derivatives](#)
 - (h) [Killing vectors and Fermi-Walker transport](#)
7. Einstein's field equations and their linear approximation
 - (a) [The field equations](#)

- (b) [The linear approximation and the Newtonian limit](#)
 - (c) [Field equations from a variational principle](#)
 - (d) [Some important examples](#)
 - (e) [Conservation laws and General remarks on the structure of the theory](#)
8. [Gravitational waves](#)
- (a) [Propagation of gravitational waves](#)
 - (b) [Propagation of gravitational waves and the energy in gravitational wave](#)
 - (c) [Emission of gravitational waves](#)
 - (d) [Emission of gravitational waves and Detection of gravitational waves](#)
9. [The Schwarzschild solution](#)
10. [Summary and outlook](#)

Relativity and Cosmology 2

1. Classical Tests of General Relativity

- (a) Gravitational redshift
- (b) Geodesics in Schwarzschild space-time A
- (c) Geodesics in Schwarzschild space-time B
- (d) Perihelion motion of Mercury
- (e) Deflection of light and Shapiro time delay
- (f) Geodetic precession
- (g) Lense-Thirring effect

2. Differential Geometry II: modern formulation

- (a) Tangent space and tangent vectors, Differentials and cotangent space, and Tensor algebra and metric
- (b) The exterior algebra and Exterior -forms
- (c) Connections and Cartan equations
- (d) Metricity, applications, integration and Maxwell and Einstein equations

3. Relativistic Stars and Black Holes

- (a) Elementary theory of degenerate stars
- (b) Interior solution of spherically symmetric stars and Energy density and binding energy
- (c) Geometry of the Schwarzschild solution
- (d) Geometry of the TOV solution and Charged Black Holes
- (e) Rotating Black Holes
 - i. Kerr solution
 - ii. Geodesics in the Kerr geometry
 - iii. Ergosphere and Penrose process
- (f) Remarks on black-hole thermodynamics and Gravitational collapse to black holes
- (g) Astrophysical black holes

4. Foundations of Cosmology

- (a) A brief history of cosmology
- (b) Structures in the Universe and the Cosmological Principle and Spaces of constant curvature: Form of the metric

- (c) Geometric realisation of spaces with constant curvature and Spacetime metric and redshift: Robertson-Walker metric
- (d) Robertson-Walker metric and Cosmological redshift
- (e) Friedmann-Lemaître equations
- (f) Distance measures
- (g) Temporal evolution of the scale factor for $\Lambda = 0$
- (h) Horizons and Solutions for $\Lambda \neq 0$

6 Gravitational Wave Astrophysics

1. Intro and GR recap [A](#) [B](#)
2. Black Holes [A](#) [B](#)
3. Lin Gravity 1 [A](#) [B](#)
4. Lin Gravity 2 [A](#)
5. Lin Gravity 3 [A](#) [B](#)
6. Binary Inspiral [A](#) [B](#)
7. Binary GWs CO Formation [A](#) [B](#)
8. Neutron Stars [A](#) [B](#)
9. Data Analysis 1 [A](#) [B](#)
10. Data Analysis 2 [A](#) [B](#)
11. BBH Observations [A](#) [B](#)
12. BNS Observations [A](#) [B](#)
13. NSBH Observations [A](#) [B](#)
14. Other GW Results [A](#) [B](#)
15. [Future of GW](#)

7 Radiative and Dynamical Processes

1. Introductory Course Overview
2. (a) Random Walks and Diffusion
(b) Brownian Motion and Langevin Equation
3. (a) Intro to Plasma; Coulomb Collisions
(b) Coulomb Collision Rates
4. (a) Kinetic Theory, Liouville's Theorem
(b) Boltzmann Collision Term; Fokker-Planck Equation
5. (a) Fluid Moments of the Boltzmann Equation
(b) Adiabatic Thermodynamics; Intro to MHD; Magnetic Pressure and Tension
(c) Ideal and Resistive MHD; The Induction Equation
6. (a) Ideal MHD Applications; Potential and Force-Free Fields
(b) Ideal MHD Applications: MHD Waves and Instabilities
7. (a) Transport Coefficients in Non-Ideal MHD
(b) Beyond MHD; Magnetic-Field Anisotropies and Ambipolar Diffusion
8. (a) Dynamics: Work, Energy, and Euler-Lagrange Theory
(b) 2-Body Keplerian Motion Derived from Lagrangians
(c) Astrophysical Applications of 2-Body Orbits
9. (a) Finish 2-Body Applications. 3-Body Problem: Roche Lobes, Hill Stability
(b) Trojan Lagrange Points, Tidal Distortions, Tidal Heating
10. (a) N-body Gravitational Dynamics; Collisional Relaxation; Gravitational Focusing
(b) N-body Collisionless Potentials; Oort Constants; the Milky Way
(c) N-body Virial Theorem; Applications in Stars and Galaxy Clusters
11. (a) Radiation Processes: Intensity, Flux, and Magnitudes
(b) Radiation Moments; Bose-Einstein Statistics; Equation of Radiative Transfer
(c) Opacity Sources; Optical Depth; LTE and Non-LTE
(d) Solving the RT Equation for a Gray Atmosphere; Exploring Its Properties
12. (a) Radiative Transfer in Non-Gray, Non-LTE, and Non-Plane-Parallel Cases

- (b) Radiative Transfer for Spectral Lines: Opacities and Einstein Relations
- (c) Spectrum Formation and Multiple Sources of Line Broadening
- (d) Ionization and Recombination; Irradiated Planetary Atmospheres
- (e) Greenhouse Effect; Radiation Pressure; Poynting-Robertson Drag

8 Special Relativity from a Mathematician's Point of View

1. 1

2. 2

3. 3

4. 4

5. 5

6. 6

9 Coronal Heating, Solar/Stellar Winds, and Space Weather

1. [Course overview: Survey of Data, Jargon, and Unanswered Questions](#)
2. Coronal Heating:
 - (a) [Origins and Scaling Laws](#)
 - (b) [Discussion and Hands-on Exercises](#)
 - (c) [Physics of Wave/Turbulence Concepts](#)
 - (d) [Physics of Reconnection/Nanoflare Concepts](#)
3. [Magnetic Field Extrapolation and Parker's \(1958\) Solar Wind](#)
4. Solar Wind:
 - (a) [Advanced Physics; Hands-on Exercises](#)
 - (b) [Evolution, Corotating Streams, and the Outer Heliosphere](#)
5. Coronal Mass Ejections: [Acceleration, Heating, and Evolution](#)
6. Space Weather:
 - (a) [The Interplanetary Space Environment](#)
 - (b) [Terrestrial Responses and Technological Impacts](#)
 - (c) [Case Studies of Historical Extreme Events](#)
 - (d) [Research to Operations and Forecasting](#)
7. The Solar/Stellar Connection: [Scalings and Exoplanet Impacts](#)

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